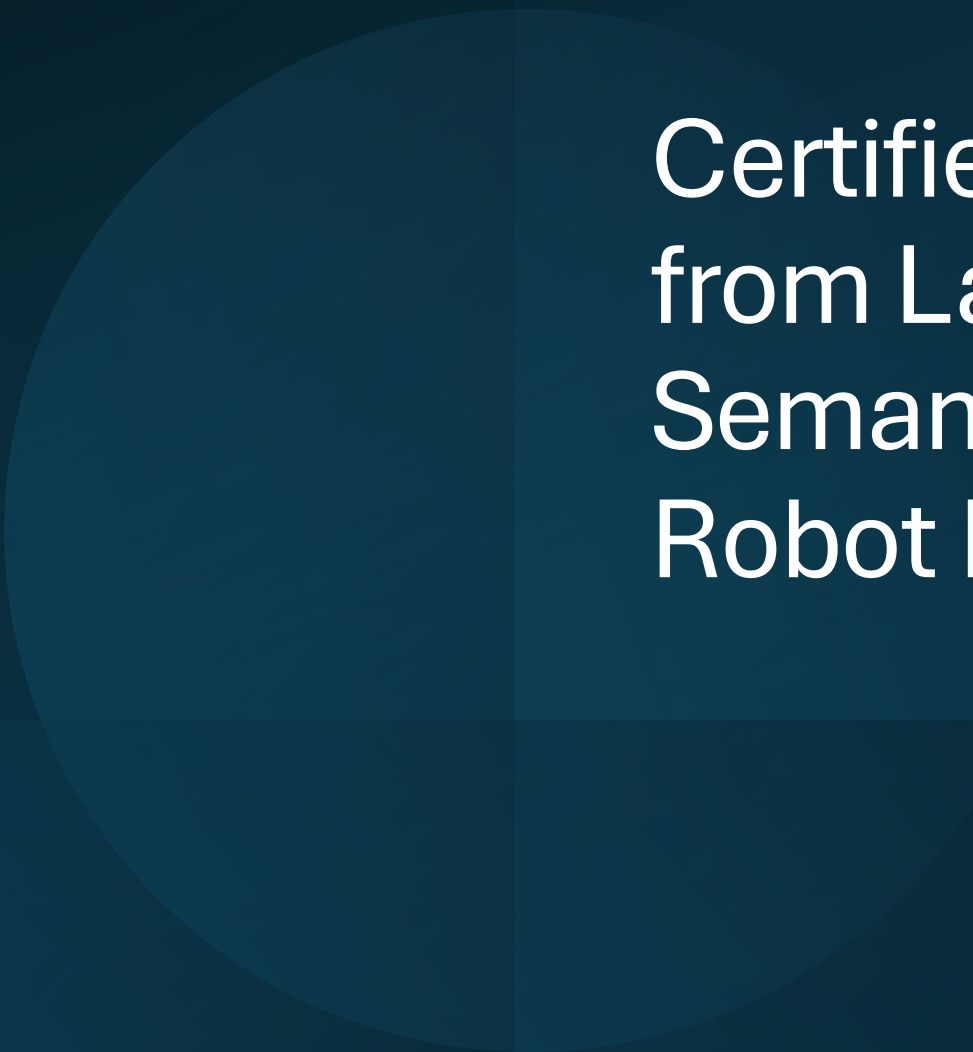
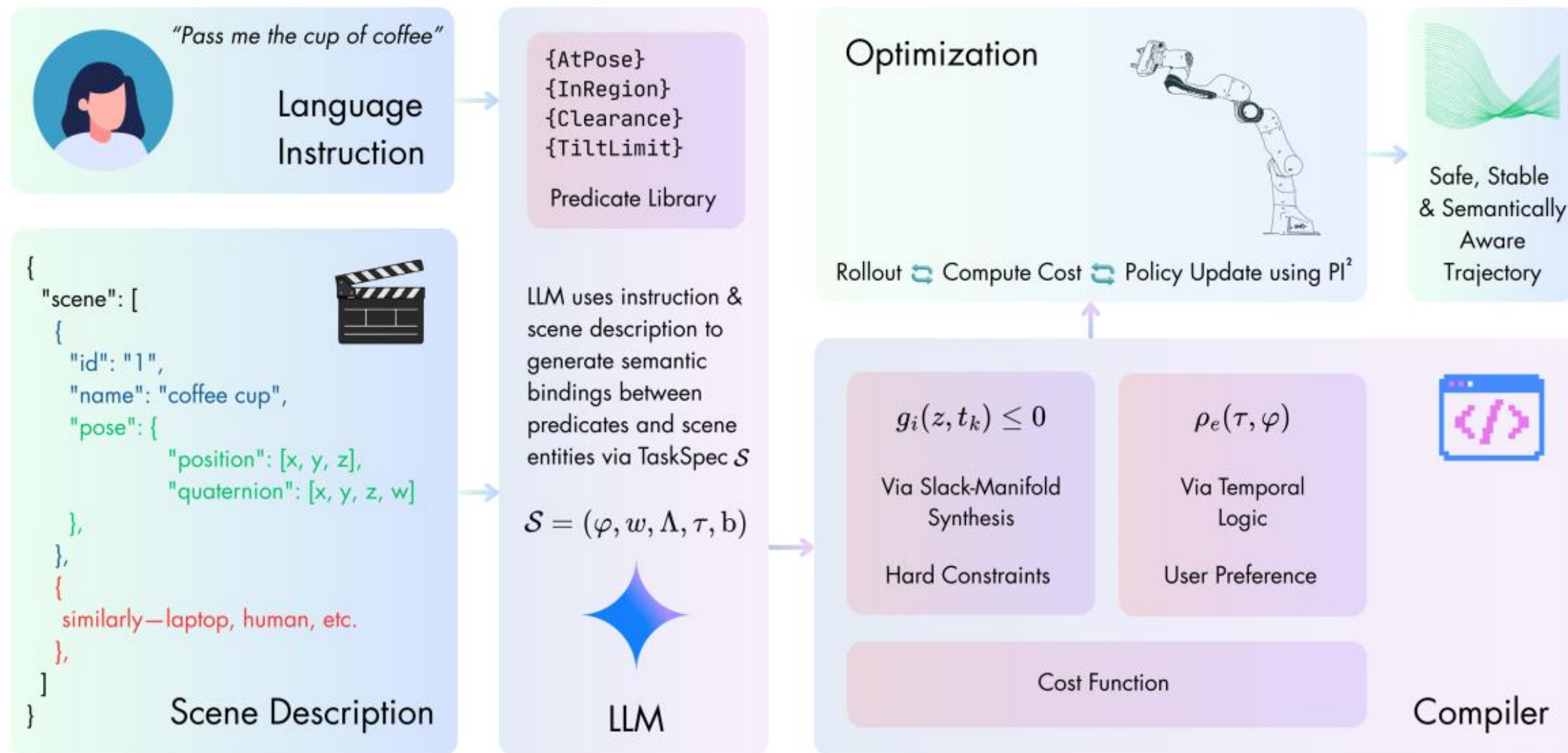


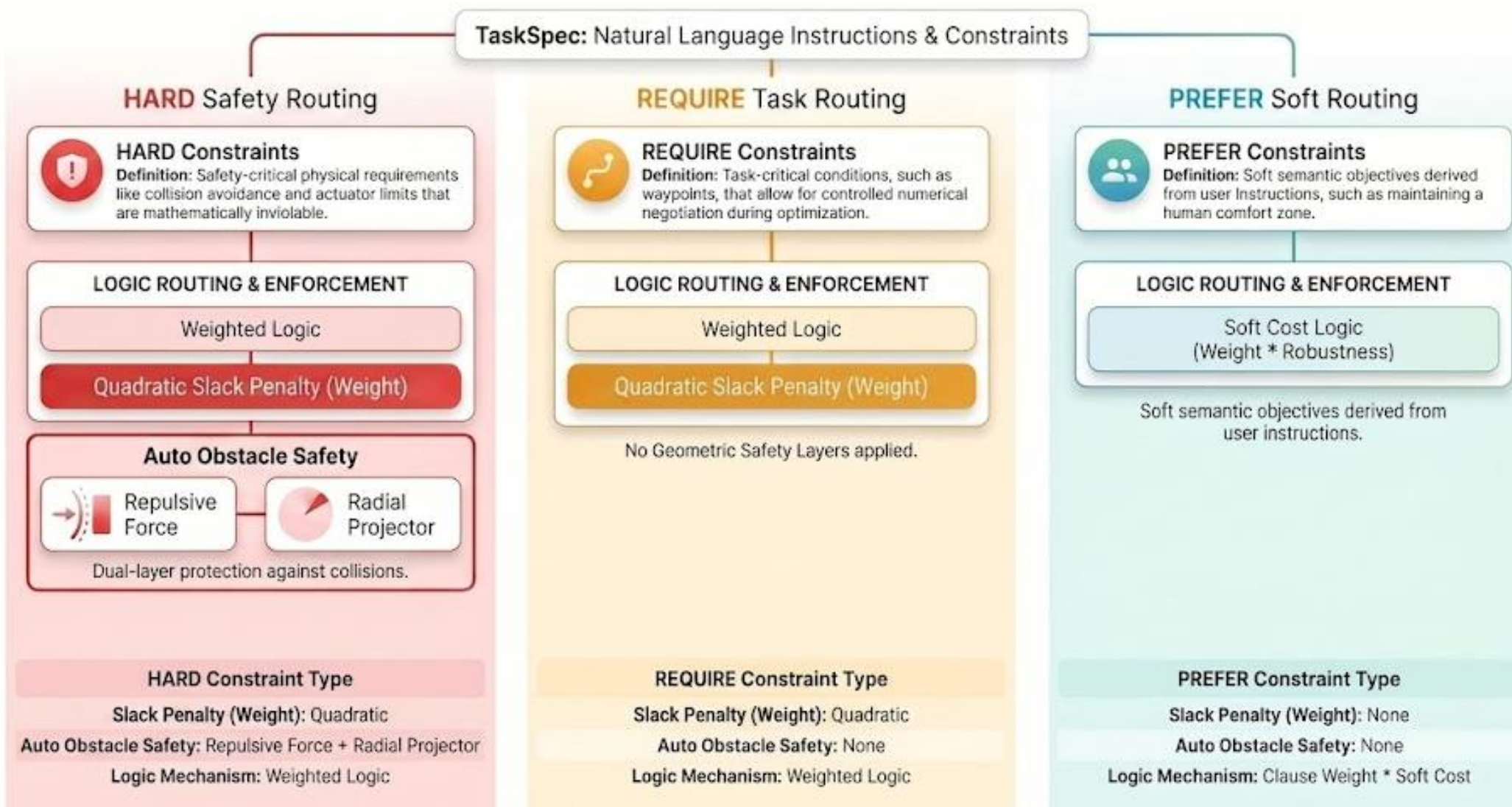


Certified Policy Synthesis from Language for Safe and Semantically-Constrained Robot Manipulation

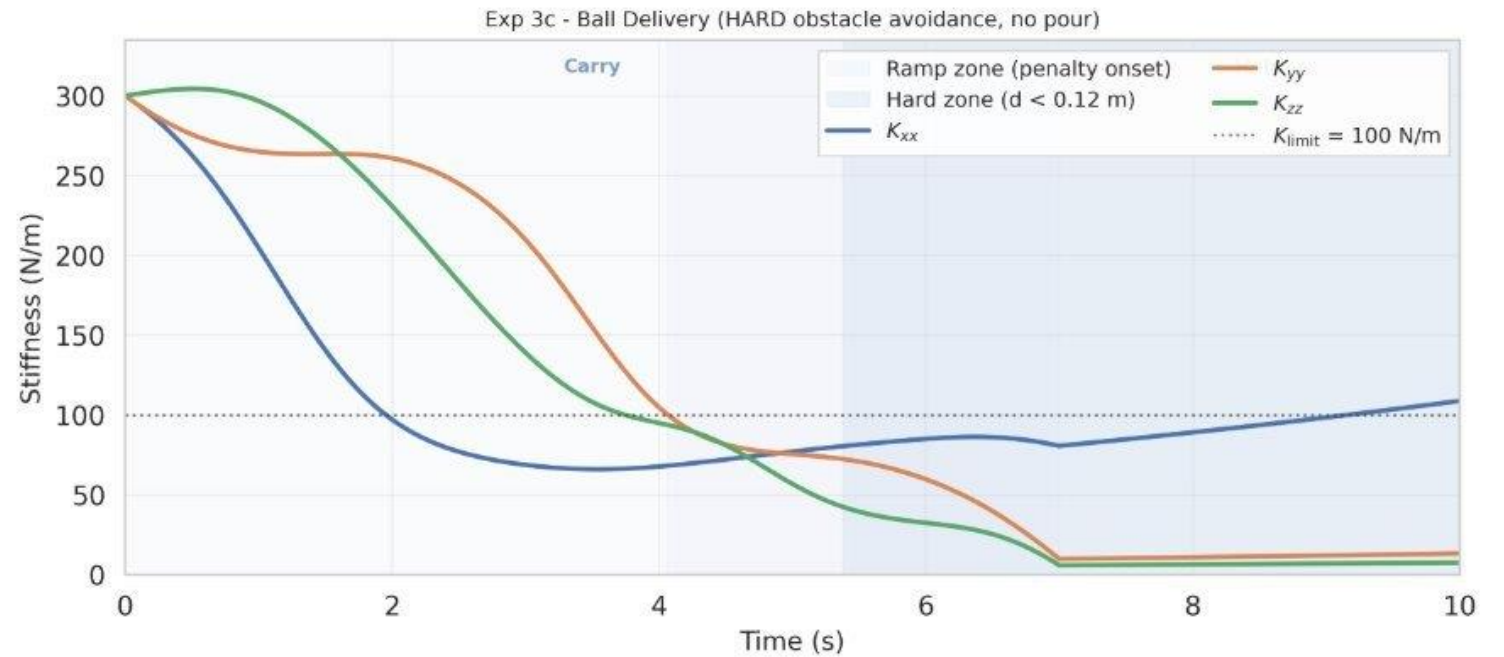
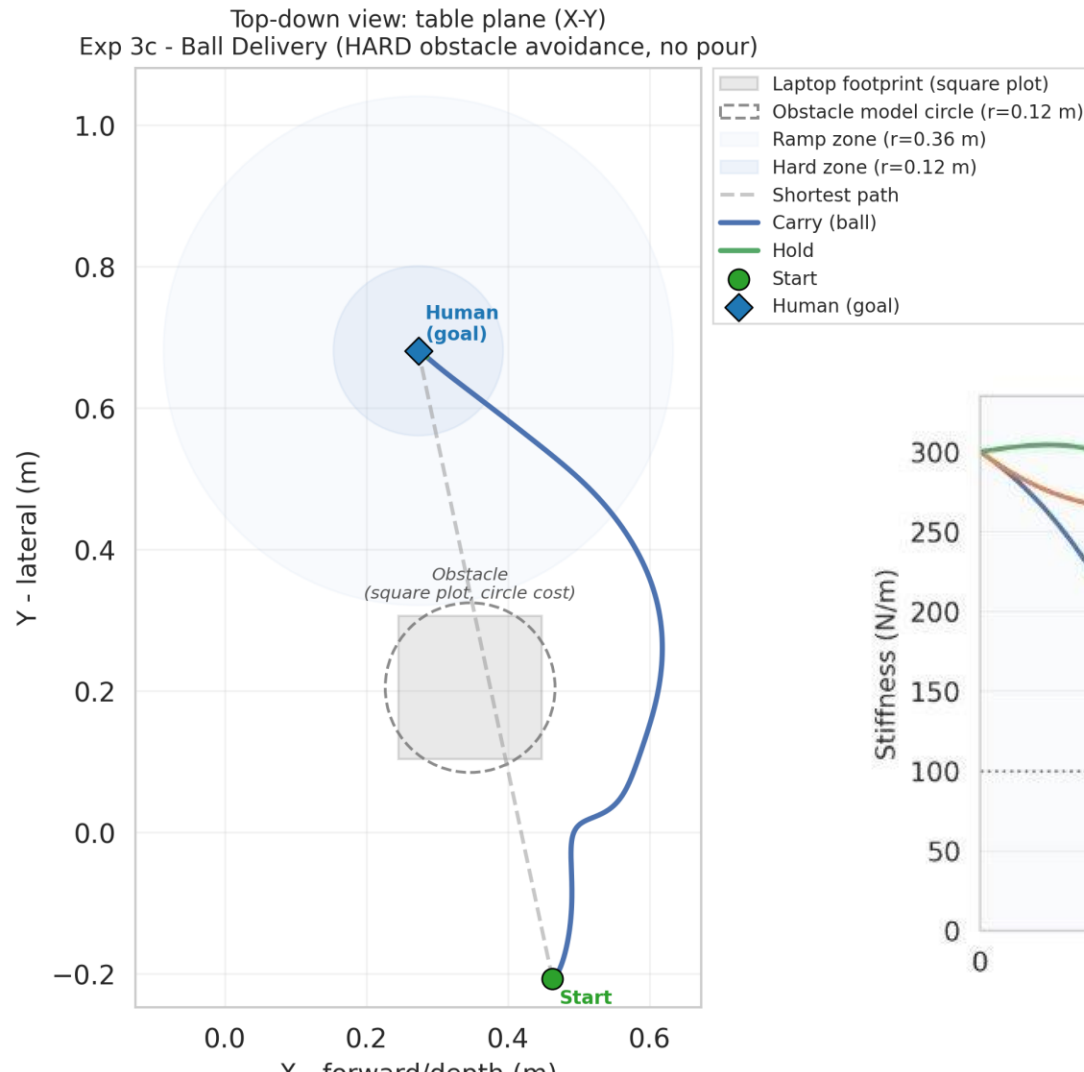
Workflow Structure



Safety Mechanism

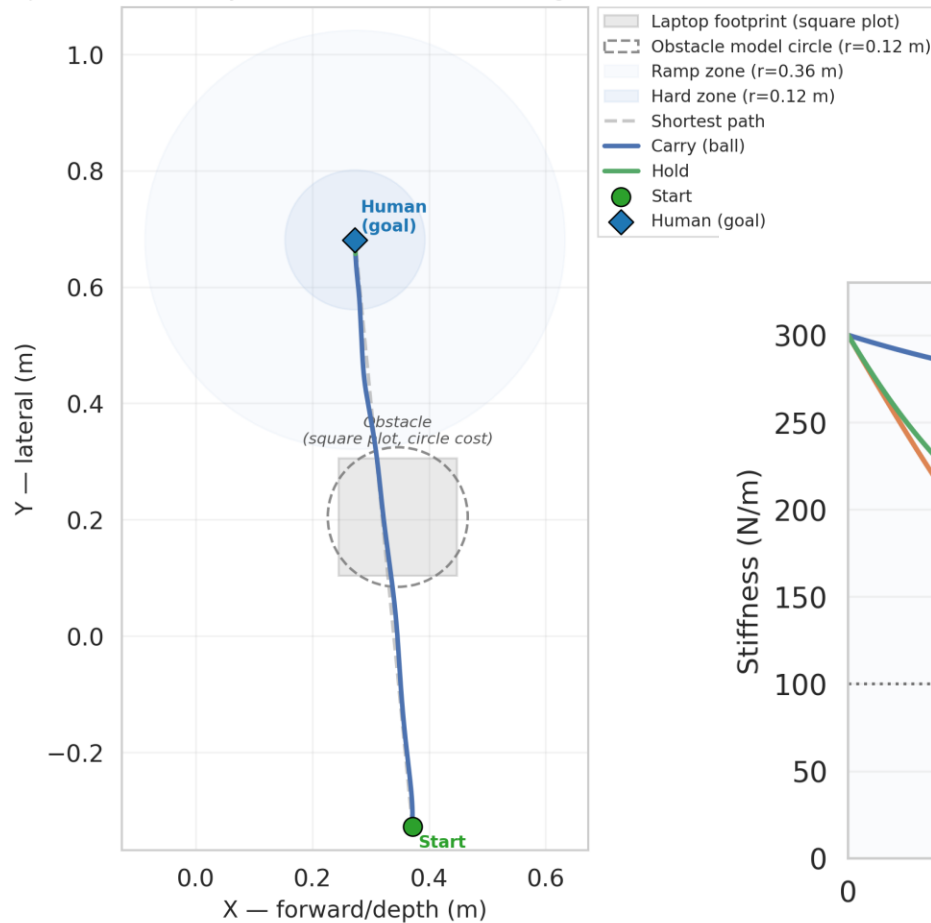


Exp 1 (a): Language-Driven Semantic Modality Selection

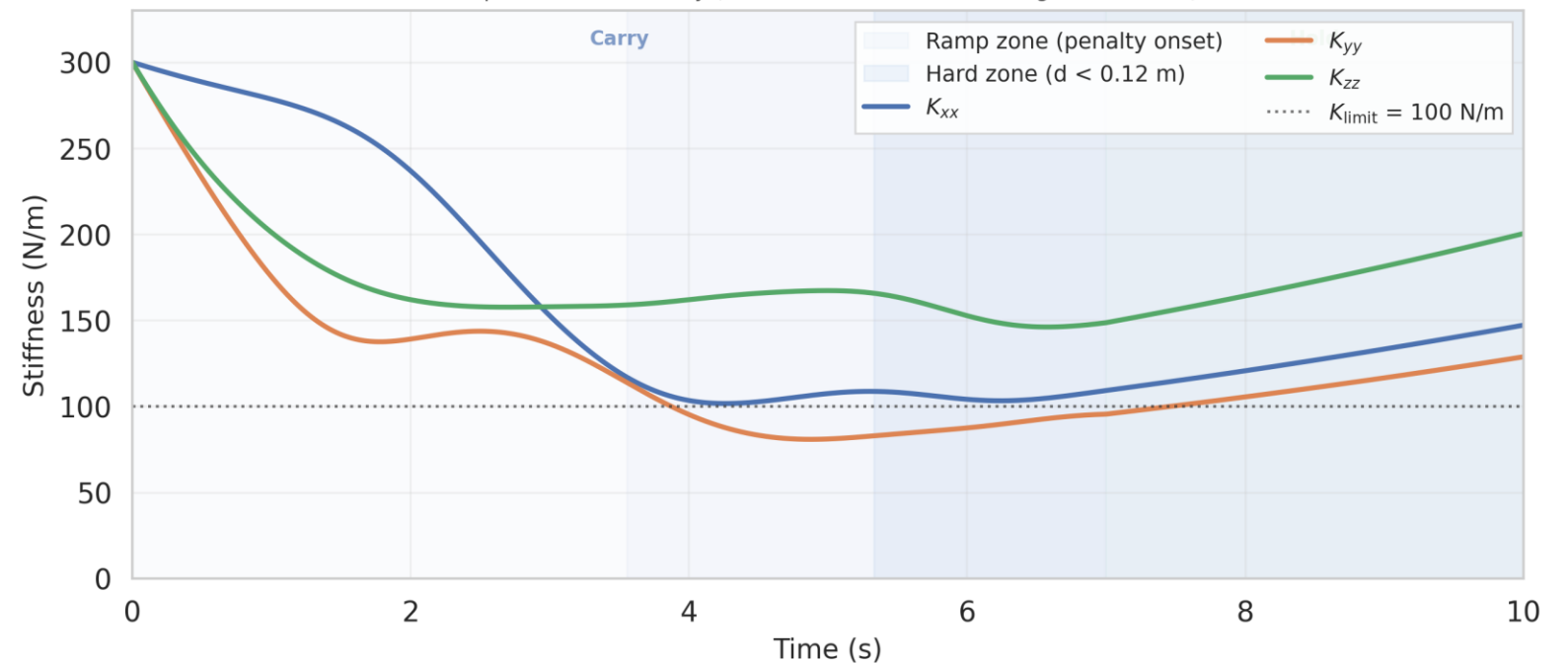


Exp 1 (b): Language-Driven Semantic Modality Selection

Top-down view: table plane (X-Y)
Exp 3b — Ball Delivery (soft obstacle avoidance, weight-controlled)

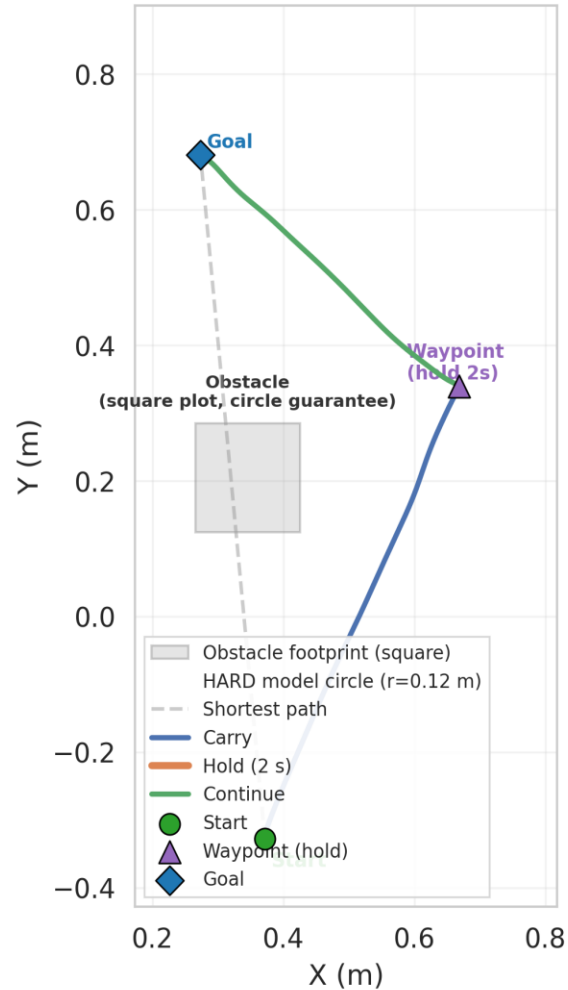


Exp 3b — Ball Delivery (soft obstacle avoidance, weight-controlled)

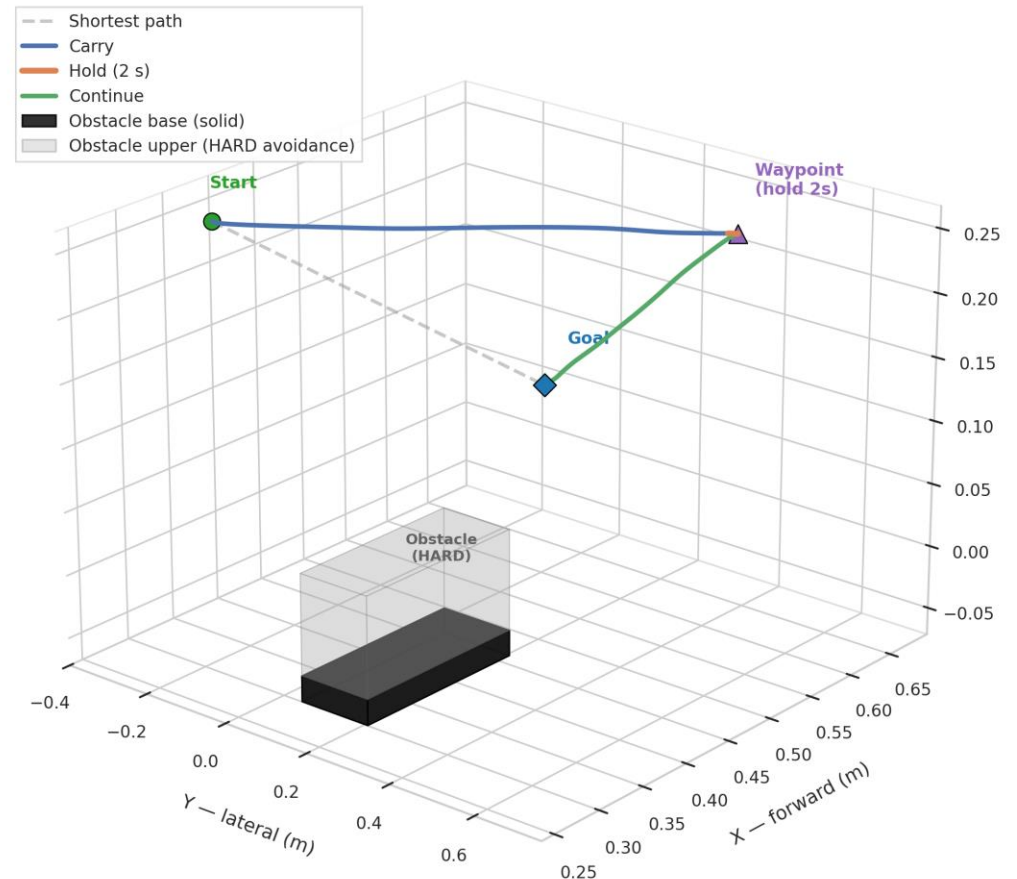


Exp 2: Multi-Phase Sequencing with Hard Obstacle Avoidance

Top-down view (X-Y)
Scene 4 — Carry → Hold (2 s) → Continue (HARD obstacle avoidance)



Scene 4 — Carry → Hold (2 s) → Continue (HARD obstacle avoidance)



Exp 3: Human-Aware Manipulation Under Temporal Trade-off

